

PHISHING AWARENESS ANALYSIS

Project 3 | Cybersecurity Internship

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1. Executive Summary

Phishing is a trick used by attackers to fool people into clicking unsafe links, sharing private information, or taking an action that puts their account or company at risk. This report covers Project 3 of the DecodeLabs Cybersecurity Internship, which focuses on Phishing Awareness Analysis. The goal is to look at sample emails and messages and find the signs that show a message is a phishing attempt.

For this report, five simulated messages were created and studied. Each message was checked for a suspicious sender, urgent or threatening wording, unsafe links, requests for personal information, and unrealistic offers. A red-flag checklist and a simple risk level (High, Medium, or Low) were used to organize the results.

2. Project Objective

The objective of this project is to analyze sample emails or messages and identify which ones are phishing attempts. This follows the internship requirement of identifying suspicious links or keywords, listing red flags, and explaining why a message is unsafe.

- Identify suspicious links and keywords in sample messages.
- Recognize common red flags found in phishing messages.
- Explain the security risk connected to each suspicious message.
- Apply basic threat-analysis and cybersecurity-awareness thinking.
- Sort each analyzed message into a risk level.

3. Project Scope

This project only looks at the text of emails and messages. No suspicious link or attachment was actually opened or visited during the analysis.

Area	Scope
Sender Analysis	Check the sender's name and email domain for anything unusual.
Content Analysis	Read the subject line, wording, and requests made in the message.
Keyword Analysis	Look for words related to urgency, threats, rewards, or verification.
URL Analysis	Study links as plain text, without clicking or visiting them.
Risk Assessment	Give each message a risk level based on how many red flags it has.
Documentation	Write down the findings and recommendations in a clear report.

4. Methodology

The following steps were followed to complete this analysis:

- Prepare simulated sample emails and messages that are safe to study.
- Check the sender address and look for a misleading or fake domain.
- Read the subject line and body for pressure, threats, or unusual requests.
- Highlight suspicious keywords that push the reader to act quickly.
- Look at the links written in the message without opening them.
- Compare each message against the red-flag checklist (Section 5).
- Give a risk level based on how many and how serious the red flags are.
- Write the final decision and explain why the message is or is not safe.

5. Phishing Red-Flag Checklist

This checklist was used to check every sample message in the same, consistent way.

No.	Red Flag	Why It Matters
1	Unknown or odd sender	The sender may not really be the person or company they claim to be.
2	Misleading domain	A look-alike domain can be used to copy a trusted brand or company.
3	Urgent language	Pushes the reader to act fast without checking the message properly.
4	Threatening language	Fear of losing an account can make people act without thinking.
5	Suspicious URL	A link may lead to a fake or dangerous website.
6	Request for sensitive info	Passwords or account details could be stolen this way.
7	Unexpected attachment	An unknown file can carry a virus or malware.
8	Unrealistic reward	A surprise prize is often used to attract victims.
9	Odd wording or errors	Poor spelling or grammar can be a sign of a fake message.
10	Demand for immediate action	Stops the reader from taking time to verify the message.

6. Sample Phishing Messages

Five simulated messages were created for this project. They copy common phishing scenarios but are safe examples made only for training.

Sample 1 — Fake Bank Alert

Sender: security@secure-bank-alert.example | Subject: URGENT: Your Account Will Be Suspended

Message: "Your account will be suspended today. Verify your account immediately using the link below: <http://secure-bank-login.example/verify>"

Sample 2 — Fake Prize Message

Sender: winner@reward-example.com | Subject: Congratulations! You Won a Prize

Message: "You have been selected to receive a \$1,000 reward. Click below to claim your prize immediately: <http://claim-reward.example/prize>"

Sample 3 — Fake Password Expiry Alert

Sender: admin@account-security.example | Subject: Password Expiration Notice

Message: "Your password will expire today. Please confirm your account information using: <http://account-check.example/login>"

Sample 4 — Suspicious Delivery Message

Sender: delivery@package-update.example | Subject: Delivery Address Confirmation Required

Message: "Your package could not be delivered because your address is incomplete. Update your delivery details using: <http://delivery-update.example/address>"

Sample 5 — Normal Training Notice (Safe Example)

Sender: training@example.edu | Subject: Internship Training Schedule

Message: "Your cybersecurity training session is scheduled for Monday at 10:00 AM. Please review the training schedule through the official student portal."

7. Detailed Message Analysis

ID	Type	Suspicious Keywords	Reason	Risk
P-01	Fake Bank Alert	Urgent; suspended; immediately	Uses urgency, threat, and a fake domain to push the reader toward a suspicious verification link.	High
P-02	Fake Prize	Congratulations; reward; claim	An unexpected prize plus urgent wording encourages an unsafe click from an unknown sender.	High
P-03	Password Alert	Expire; today; confirm	Pressures the reader to confirm account details through a suspicious login page.	Medium–High
P-04	Delivery Alert	Required; update; details	Requests personal information through an unfamiliar delivery link.	Medium
P-05	Training Notice	Schedule; training; portal	Informational message with no urgency, threats, or requests for private data.	Low

8. Suspicious Keyword Analysis

Certain words appeared often in the unsafe sample messages. On their own these words are not always dangerous, but combined with other red flags they are a strong warning sign.

Keyword	Potential Warning Sign
Urgent / Immediately	Creates pressure to act quickly without checking the message.
Suspended	Creates fear of losing access to an account.
Verify / Confirm	May lead to a fake login or verification page.
Password	Points to a sensitive login target.
Congratulations / Claim	Often used to introduce a fake reward or prize.
Required	Pushes the reader to complete an action right away.
Expire	Creates a false sense of a running-out deadline.
Click	Directly pushes the reader toward an unsafe link.

9. URL Analysis

Links found in the sample messages were studied as plain text only, without visiting them, and were checked for misleading wording such as "secure" or "login."

Example Link	Concern
http://secure-bank-login.example/verify	Uses banking wording but the domain is not a real, trusted bank.
http://claim-reward.example/prize	Uses reward wording to lure the reader into an unsafe claim page.
http://account-check.example/login	Copies login wording and is likely built to steal account details.
http://delivery-update.example/address	Requests personal address details through an unfamiliar domain.

Shortened links (like bit.ly-style URLs) can also hide a dangerous destination behind a short, harmless-looking address, so any shortened or unfamiliar link should always be checked before use.

10. Risk Classification

Every message was placed into a risk category based on how many and how serious its red flags were.

Risk Level	General Meaning
High	Several strong red flags together, such as a suspicious link, urgency, a fake domain, and a request for private information.
Medium	Some suspicious signs are present, and the message should be checked before trusting it.
Low	No major red flags were found; the message looks safe and normal.

Out of the five sample messages studied: 2 were rated High risk, 1 was rated Medium–High, 1 was rated Medium, and 1 (the normal training notice) was rated Low.

11. Overall Findings

- Phishing messages usually combine more than one trick at the same time, not just one warning sign.
- Urgency and threats (like account suspension) were the most common way to pressure the reader.
- Suspicious links appeared in almost every unsafe sample and pointed to unfamiliar domains.
- Several fake messages asked directly for account or personal information.
- A fake reward or prize was used at least once to attract the reader's attention.
- Checking the sender's domain was one of the fastest ways to catch a fake message.

12. Recommendations

- Always check the sender's full email address and domain before trusting a message.
- Never click a suspicious link or open an unexpected attachment.
- Never share a password, OTP, or other private details through a link in an email.
- Read links carefully and confirm the true destination through a trusted, separate channel.
- Use a red-flag checklist, like the one in Section 5, to check messages the same way every time.
- Report any suspected phishing message to the IT or security team instead of just deleting it.
- Take part in regular phishing-awareness training to stay updated on new tricks.

13. Limitations

- This project used simulated, safe sample messages rather than real company email traffic.
- No suspicious link was actually opened or clicked during the analysis.
- The focus of this project is awareness and manual identification, not automatic detection.
- Risk levels are based on the training examples used here and are not a guarantee of safety in every real situation.

14. Future Enhancements

This manual analysis process could be extended into an automated phishing-detection tool. A future version could scan sender details, message content, links, and keywords automatically and produce a risk score.

- Automatic checking of a link's reputation.

- Using machine learning to classify phishing messages.
- Automatic risk scoring instead of manual review.
- Connecting the tool with existing email security systems.
- Building a simple dashboard to track and report phishing attempts.

15. Conclusion

This project studied simulated emails and messages to find common phishing warning signs such as suspicious links, urgent wording, fake domains, and requests for private information. Using a red-flag checklist and a simple risk level made it much easier to organize the findings and judge how dangerous each message was.

Overall, this project gave practical, hands-on experience in phishing awareness, threat identification, and basic cybersecurity thinking, and it builds a strong foundation before moving on to more technical security topics.

16. References

- [1] DecodeLabs. Industrial Training Kit — Cyber Security Project 3: Phishing Awareness Analysis, Batch 2026.
- [2] Federal Trade Commission. "How To Recognize and Avoid Phishing Scams."
- [3] Microsoft. "Protect Yourself from Phishing."
- [4] NIST. "Can You Spot a Phish?", 2022.
- [5] Google. "Avoid & Report Phishing Emails."